

8. RANK CORRELATION

8.1 Introduction

When a group of individuals are arranged according to their degree of possession of a character, they are said to be ranked. The ordinal number of an individual in the arrangement is called its *rank* and the arrangement as a whole is called a *ranking*. When there are two series of ranks for the same set of individuals, corresponding to two different characters or two judges assigning ranks for the same character, one may be interested to know if the two series are associated. The association between two series of ranks for the same set of individuals is called *rank correlation*.

8.2 Spearman's rank correlation coefficient

Case (a) : When there are no ties

Suppose we have n individuals and their ranks in the two series are $u_1, u_2, \dots, u_n$ and $v_1, v_2, \dots, v_n$. When there are no ties, u 's and v 's are some permutations of the first n natural numbers. Let $d_i = u_i - v_i$, $i = 1(1)n$. Then Spearman's rank correlation coefficient, denoted by r_R , is given by

$$r_R = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}.$$

When there is perfect agreement between the two series, association is said to be positively perfect, and in that case $u_i = v_i$, for each i , and hence $\sum d_i^2 = 0$. So $r_R = 1$.

Again, when there is perfect disagreement between the two series, association is said to be negatively perfect, and in that case $v_i = n - u_i + 1$, for each i , and hence

$$\begin{aligned} \sum d_i^2 &= \sum (2u_i - n - 1)^2 = 4 \sum u_i^2 - 4(n+1) \sum u_i + n(n+1)^2 \\ &= 4 \cdot \frac{n(n+1)(2n+1)}{6} - 4(n+1) \cdot \frac{n(n+1)}{2} + n(n+1)^2 \\ &= \frac{2}{3} n(n+1)(2n+1) - n(n+1)^2 = \frac{1}{3} n(n^2 - 1) \\ \therefore r_R &= 1 - \frac{6n(n^2 - 1)}{3} \cdot \frac{1}{n(n^2 - 1)} = 1 - 2 = -1. \end{aligned}$$

Spearman's rank correlation coefficient can be derived as the simple product-moment correlation coefficient taking u_i and v_i as variate values.

$$\bar{u} = \frac{1}{n} \sum_i u_i = \frac{1}{n} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2}$$

$$\begin{aligned} \text{and } s_u^2 &= \frac{1}{n} \sum_i (u_i - \bar{u})^2 = \frac{1}{n} \sum_i u_i^2 - \bar{u}^2 = \frac{1}{n} \cdot \frac{n(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 \\ &= \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{n^2-1}{12}. \end{aligned}$$

$$\text{Similarly, } \bar{v} = \frac{n+1}{2} \text{ and } s_v^2 = \frac{n^2-1}{12}.$$

$$\begin{aligned} \text{Again, } \frac{1}{n} \sum_i di^2 &= \frac{1}{n} \sum_i (u_i - v_i)^2 = \frac{1}{n} \sum_i \{(u_i - \bar{u}) - (v_i - \bar{v})\}^2, \text{ (since } \bar{u} = \bar{v}). \\ &= \frac{1}{n} \sum_i (u_i - \bar{u})^2 + \frac{1}{n} \sum_i (v_i - \bar{v})^2 - \frac{2}{n} \sum_i (u_i - \bar{u})(v_i - \bar{v}) \\ &= s_u^2 + s_v^2 - 2 \operatorname{cov}(u, v) = \frac{n^2-1}{6} - 2 \operatorname{cov}(u, v). \end{aligned}$$

$$\therefore \operatorname{cov}(u, v) = \frac{n^2-1}{12} - \frac{1}{2n} \sum_i di^2.$$

$$\text{So, } r_{uv} = \frac{\operatorname{cov}(u, v)}{s_u s_v} = \frac{\frac{n^2-1}{12} - \frac{1}{2n} \sum_i di^2}{\frac{n^2-1}{12}} = 1 - \frac{6 \sum_i di^2}{n(n^2-1)} = r_R.$$

Case (b) : When there are ties

In practical application of ranking methods, we find situations when two or more individuals are so similar that no preference can be made between them. These individuals are then said to be tied. The arrangement of students by reference to examination marks is a familiar source of ties. If k individuals are allotted the same rank, then we have a tie of length k . The method adopted for allocating rank-numbers to tied individuals is to average the ranks which they would possess if they were distinguishable. When the k individuals having same rank follow r other individuals in the ranking, then each of them is given the rank

$$\frac{1}{k} [(r+1) + (r+2) + (r+3) \dots + (r+k)] = \frac{1}{k} \left[rk + \frac{k(k+1)}{2} \right] = r + \frac{k+1}{2}.$$

If they were untied, the sum of squares of their ranks would be

$$\begin{aligned} S_1 &= (r+1)^2 + (r+2)^2 + (r+3)^2 + \dots + (r+k)^2 \\ &= kr^2 + 2r(1+2+3+\dots+k) + (1^2+2^2+3^2+\dots+k^2) \\ &= kr^2 + k(k+1)r + \frac{1}{6}k(k+1)(2k+1), \end{aligned}$$

whereas, the sum of squares of the tied ranks is

$$S_2 = k \left[r + \frac{k+1}{2} \right]^2 = k \left[r^2 + (k+1)r + \frac{(k+1)^2}{4} \right] = kr^2 + k(k+1)r + \frac{1}{4}k(k+1)^2$$

Now,

$$S_1 - S_2 = k(k+1) \left(\frac{2k+1}{6} - \frac{k+1}{4} \right) = \frac{k(k+1)(k-1)}{12} = \frac{1}{12}(k^3 - k).$$

Since sum of squares of ranks decreases by $\frac{1}{12}(k^3 - k)$, but mean remains unchanged, the variance of the series decreases by $\frac{1}{12n}(k^3 - k)$.

Suppose there are l ties of length $k_1, k_2, \dots, k_l$ in the first ranking and m ties of length $k'_1, k'_2, \dots, k'_m$ in the second ranking. Since effect of tying different sets is additive, the variances would be

$$S_u^2 = \frac{n^2 - 1}{12} - T_u, \text{ and } S_v^2 = \frac{n^2 - 1}{12} - T_v,$$

where $T_u = \frac{1}{12n} \sum_{i=1}^l (k_i^3 - k_i)$ and $T_v = \frac{1}{12n} \sum_{i=1}^m (k_i'^3 - k_i')$.

Again, since $\frac{1}{n} \sum_{i=1}^n d_i^2 = s_u^2 + s_v^2 - 2 \text{cov}(u, v)$, we have, in the case of tied ranks,

$$\text{cov}(u, v) = \frac{n^2 - 1}{12} - \frac{T_u + T_v}{2} - \frac{1}{2n} \sum_{i=1}^n d_i^2$$

So, in the case of tied ranks, Spearman's rank correlation coefficient is

$$r_R = \frac{\frac{n^2 - 1}{12} - \frac{T_u + T_v}{2} - \frac{1}{2n} \sum_{i=1}^n d_i^2}{\sqrt{\frac{n^2 - 1}{12} - T_u} \sqrt{\frac{n^2 - 1}{12} - T_v}}$$

In case of perfect agreement between the two series of ranks, we have, for each i , $u_i = v_i$, and hence $d_i = 0$, $T_u = T_v$ and $S_u = S_v$.

$$\therefore r_R = 1.$$

Alternatively, here the relationship between the variables u and v being represented exactly by a straight line with a positive slope, $r_R = 1$.

In case of perfect disagreement, $v_i = n - u_i + 1$, i.e. the relationship between the variables u and v is represented exactly by a straight line with negative slope.

$$\therefore r_R = -1.$$

8.3 Kendall's rank correlation coefficient

Case (a) : When there are no ties

We consider each possible pair of individuals and the order in which they occur in the two rankings. We allot the pair a score of +1 if they occur in the same order in the two series, and a score of -1 if they occur in reverse order. A score is thus obtained for each of the $\binom{n}{2} = \frac{1}{2}n(n-1)$ possible pairs of individuals. Kendall's rank correlation coefficient τ is then given by

$$\tau = \frac{\text{total score}}{\text{maximum possible total score}} = \frac{\text{total score}}{\frac{1}{2}n(n-1)}$$

If P and Q denote respectively the total of positive scores and total of negative scores, we have

$$\begin{aligned}\tau &= \frac{P-Q}{\frac{1}{2}n(n-1)} = 1 - \frac{2Q}{\frac{1}{2}n(n-1)}, \quad \left[\because P+Q = \frac{n(n-1)}{2} \right] \\ &= \frac{2P}{\frac{1}{2}n(n-1)} - 1.\end{aligned}$$

This shows that for computation of τ , when there is no tie, we need to determine only P (or only Q).

The determination of total score (or equivalently of P or of Q) becomes easy if we rearrange the ranks in one series in the natural order and consider the corresponding ranking in the other series. The contribution to P will arise only from pairs in the second series which are in the natural order. These are all we need to count.

It is to be noted that, in case of perfect agreement between the two rankings, the score from each of the possible $\binom{n}{2}$ pairs is +1, and hence

$$\text{total score} = \text{maximum possible score} = \frac{1}{2}n(n-1).$$

$$\therefore \tau = +1$$

Again, for perfect disagreement, the score from each possible pair is -1. So,

$$\text{total score} = \text{minimum possible score} = -\frac{1}{2}n(n-1).$$

$$\therefore \tau = -1.$$

Kendall's rank correlation coefficient τ can be regarded as a product-moment correlation coefficient. Let u_i and v_i denote the ranks of the i th individual in the two rankings,

$i = 1, 2, \dots, n$. Then, for the pair of individuals (i, j) , where $i < j$, we define two quantities a_{ij} and b_{ij} such that

$$a_{ij} = \begin{cases} +1 & \text{if } u_i < u_j \\ -1 & \text{if } u_i > u_j \end{cases} \quad \text{and} \quad b_{ij} = \begin{cases} +1 & \text{if } v_i < v_j \\ -1 & \text{if } v_i > v_j \end{cases}$$

So, it follows that

$$\tau = \frac{\sum a_{ij} b_{ij}}{\sqrt{\sum a_{ij}^2} \sqrt{\sum b_{ij}^2}},$$

where each summation is taken over all possible pairs of individuals (i, j) , with $i < j$.

Case (b) : *When there are ties*

In this case we have to take

$$a_{ij} = 0 \text{ if } u_i = u_j \text{ and } b_{ij} = 0 \text{ if } v_i = v_j.$$

Since $a_{ij} \times b_{ij} = 0$ when either a_{ij} is zero or b_{ij} is zero or both a_{ij} and b_{ij} are zero, a tie of length k in a series will reduce the score by $\frac{1}{2}k(k-1)$. So if there be l ties of length $k_1, k_2, \dots, k_l$ in the first ranking, we have,

$$\sum a_{ij}^2 = \frac{1}{2}n(n-1) - T'_u,$$

$$\text{where } T'_u = \frac{1}{2} \sum_{i=1}^l k_i(k_i - 1).$$

Similarly, for m ties of length $k'_1, k'_2, \dots, k'_m$ in the second ranking,

$$\sum b_{ij}^2 = \frac{1}{2}n(n-1) - T'_v,$$

$$\text{where } T'_v = \frac{1}{2} \sum_{i=1}^m k'_i(k'_i - 1).$$

The computation of total score will become simple by rearranging the ranks in one series in the natural order, but for the score corresponding to any pair of individuals (i, j) , with $i < j$, we have to consider the values of both a_{ij} and b_{ij} .

The formula for kendall's rank correlation coefficient τ , in the case of tied rank, then becomes

$$\tau = \frac{\text{Total score}}{\sqrt{\frac{1}{2}n(n-1) - T'_u} \sqrt{\frac{1}{2}n(n-1) - T'_v}}.$$

8.4 Uses of rank correlation

The different uses of rank correlation are as follows :

(i) Rank correlation is used to measure the extent of association that may exist between two series of ranks for the same set of individuals.

(ii) Attributes cannot be measured, but individuals can be ranked according to their degree of possession of an attribute. Hence, rank correlation is used to see if two attributes are correlated or not.

(iii) When two judges assign ranks to the same group of individuals according to a character, one may be interested to know the extent of agreement between the judges. Rank correlation serves that purpose.

(iv) Even when measurements on two variables are available, rank correlation is sometimes used as a substitute for the simple correlation because it is very easy and quick to calculate.

Illustration 8.1

Ten boys are ranked according to their ability in Statistics and sports. Find Spearman's and Kendall's rank correlation coefficients.

Boy :	A	B	C	D	E	F	G	H	I	J
Statistics :	7	4	3	10	6	2	9	8	1	5
Sports :	5	7	3	10	1	9	6	2	8	4

The differences d_i between the two series of ranks for the 10 boys are 2, -3, 0, 0, 5, -7, 3, 6, -7, 1.

$$\therefore \sum d_i^2 = 4 + 9 + 0 + 0 + 25 + 49 + 9 + 36 + 49 + 1 = 182.$$

So, Spearman's coefficient is

$$r_R = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} = 1 - \frac{6 \times 182}{990} = 1 - 1.103 = -0.103.$$

To calculate Kendall's coefficient, we rearrange the ranking of statistics in the natural order, and thus :

Boy :	I	F	C	B	J	E	A	H	G	D
Statistics :	1	2	3	4	5	6	7	8	9	10
Sports :	8	9	3	7	4	1	5	2	6	10

Denoting the number of pairs that have ranks in the natural order in the second series by P, we have

$$P = 2 + 1 + 5 + 1 + 3 + 4 + 2 + 2 + 1 = 21.$$

So, Kendall's coefficient is

$$\tau = \frac{2P}{\binom{10}{2}} - 1 = \frac{42}{45} - 1 = -\frac{1}{15} = -0.07.$$

In both cases, the coefficient is small ; so there is very little correlation between the two rankings.

Illustration 8.2

In an examination 9 students obtained the following marks in English and Mathematics. Find Spearman's and Kendall's rank correlation coefficients.

Students (Roll Nos.) :	1	2	3	4	5	6	7	8	9
Marks in Eng. :	45	60	32	45	32	32	58	56	47
Marks in Maths :	51	51	38	54	54	38	62	58	38

Let us assign ranks to the students according to their marks.

Roll Nos. :	1	2	3	4	5	6	7	8	9
Eng. :	5 $\frac{1}{2}$	1	8	5 $\frac{1}{2}$	8	8	2	3	4
Maths. :	5 $\frac{1}{2}$	5 $\frac{1}{2}$	8	3 $\frac{1}{2}$	3 $\frac{1}{2}$	8	1	2	8

In the first ranking, there are two ties of length 2 and 3.

$$\therefore T_u = \frac{1}{12n} [(2^3 - 2) + (3^3 - 3)] = \frac{30}{12 \times 9} = \frac{5}{18} = 0.2778.$$

In the second ranking, there are three ties of length 2, 3 and 2.

$$\therefore T_v = \frac{1}{12n} [(2^3 - 2) + (3^3 - 3) + (2^3 - 2)] = \frac{36}{12 \times 9} = \frac{1}{3} = 0.3333.$$

$$\text{Here } \frac{n^2 - 1}{12} = \frac{80}{12} = \frac{20}{3} = 6.6667$$

$$\begin{aligned} \text{and } \frac{1}{12n} \sum d_i^2 &= \frac{1}{2 \times 9} \left[0 + \frac{81}{4} + 0 + 4 + \frac{81}{4} + 0 + 1 + 1 + 16 \right] \\ &= \frac{1}{18} \times \frac{250}{4} = \frac{125}{36} = 3.4722. \end{aligned}$$

So, Spearman's rank correlation coefficient is

$$\begin{aligned} r_R &= \frac{\frac{n^2 - 1}{12} - \frac{T_u + T_v}{2} - \frac{1}{2n} \sum d_i^2}{\sqrt{\frac{n^2 - 1}{12} - T_u} \sqrt{\frac{n^2 - 1}{12} - T_v}} = \frac{6.6667 - \frac{1}{2}(0.2778 + 0.3333) - 3.4722}{\sqrt{6.6667 - 0.2778} \sqrt{6.6667 - 0.3333}} \\ &= \frac{2.88895}{\sqrt{40.46346}} = 0.45. \end{aligned}$$

For calculating Kendall's rank correlation coefficient, we rearrange the ranking for English in the natural order. This gives :

Rolls Nos. :	2	7	8	9	1	4	3	5	6
Eng. :	1	2	3	4	$5\frac{1}{2}$	$5\frac{1}{2}$	8	8	8
Maths :	$5\frac{1}{2}$	1	2	8	$5\frac{1}{2}$	$3\frac{1}{2}$	8	$3\frac{1}{2}$	8

Total score is

$$(3 - 4) + (7 - 0) + (6 - 0) + (0 - 3) + (2 - 1) + (2 - 0) - 0 = 12.$$

As there are two ties of length 2 and 3 in English,

$$T'_u = \frac{1}{2}(2 \times 1 + 3 \times 2) = 4$$

In the second ranking, there are three ties of length 2, 3 and 2

$$\therefore T'_v = \frac{1}{2}(2 \times 1 + 3 \times 2 + 2 \times 1) = 5.$$

Here $\frac{n(n-1)}{2} = 36.$

$\therefore$ Kendall's rank correlation coefficient is

$$\tau = \frac{\text{Total score}}{\sqrt{\frac{n(n-1)}{2} - T'_u} \sqrt{\frac{n(n-1)}{2} - T'_v}} = \frac{12}{\sqrt{36-4} \sqrt{36-5}} = \frac{12}{\sqrt{32 \times 31}} = 0.38.$$

EXERCISES

1. What is rank correlation ? State its uses.
2. Define Spearman's rank correlation coefficient (r_R) and derive its formula when there are no ties. How should it be modified for tied ranks ?
3. Discuss how you can find Kendall's rank correlation coefficient (τ) when (i) there are no ties, and (ii) there are ties.
4. Prove that both Spearman's r_R and Kendall's τ lie between -1 and $+1$. Interpret the marginal cases.
5. Write notes on :
 - (a) Rank correlation
 - (b) Rank correlation coefficient
 - (c) Measures of rank correlation without ties.

6. A group of ten workers of a factory is ranked according to their efficiency by two different judges as follows :

Worker	:	A	B	C	D	E	F	G	H	I	J
Judge I	:	4	8	6	7	1	3	2	5	10	9
Judge II	:	3	9	6	5	1	2	4	7	8	10

Compute the coefficient of rank correlation by Spearman's or Kendall's formula and state what you learn from the result.

7. Ranks of 12 students according to their proficiency in debate and recitation are given below. Find Spearman's and Kendall's rank correlation coefficients.

Students (Roll Nos.):	1	2	3	4	5	6	7	8	9	10	11	12
Ranks in debate :	6	2	8½	4	11	12	5	1	3	8½	7	10
Ranks in recitation :	5½	2	9	4	10½	10½	5½	2	2	7	8	12

8. 9 competitors on paper presentation were marked by the two judges A and B as follows:

Competitors	:	1	2	3	4	5	6	7	8	9
Marks by Judge A	:	45	60	32	45	32	32	58	56	47
Marks by Judge B	:	51	51	38	54	54	38	62	58	38

Find Spearman's rank correlation coefficient.

MULTIPLE CHOICE QUESTIONS

- 9.(i) Spearman's rank correlation coefficient can not exceed

(a) -1 (b) 0 (c) 1 (d) none of these.

- (ii) If $n = 10$, $\sum d^2 = 33$, then Spearman's rank correlation coefficient is equal to

(a) 0.4 (b) 0.8 (c) 0.6 (d) 0.3.

- (iii) Spearman's rank correlation coefficient between the ranks of 15 individuals given by 2 judges is -1. If for a particular individual the rank given by second judge be 9, then rank of that individual by first judge will be

(a) 9 (b) 6 (c) 7 (d) 5.

- (iv) The least value of Kendall's rank correlation coefficient's is

(a) 0 (b) -2 (c) 1 (d) -1.

ANSWERS

6. $r_R = 0.88$, $\tau = 0.69$; there is good agreement between the two rankings.
 7. $r_R = 0.96$, $\tau = 0.86$. 9. (i) (c); (ii) (b); (iii) (c); (iv) (d).